# Quadratic Equations

Formulas rendered by code — KaTeX & MathJax, never image-gen

Subject Mathematics | Grade Grade 8 | Topic Quadratic Equations

## Lesson Learning Outcomes

2 min

By the end of the lesson, the learner should be able to:

a) Recall the standard form of a quadratic equation.

b) Solve a quadratic equation using the quadratic formula.

c) Apply related formulas to real-world problems.

## Key formulas

STANDARD FORM

$$ax^2 + bx + c = 0$$

QUADRATIC FORMULA

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

DISCRIMINANT

$$\Delta = b^2 - 4ac$$

AREA OF A CIRCLE

$$A = \pi r^2$$

PYTHAGORAS' THEOREM

$$a^2 + b^2 = c^2$$

## Same formula via the MathJax engine (SVG)

QUADRATIC FORMULA — MATHJAX TO SELF-CONTAINED SVG

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

## Worked example

**Solve $x^2 - 5x + 6 = 0$. Here $a = 1$, $b = -5$, $c = 6$, so the discriminant is $\Delta = 25 - 24 = 1$, and the quadratic formula gives $x = 2$ or $x = 3$.**

## Assessment Questions

5 min

a) Write the quadratic formula from memory.

b) Solve $x^2 - 7x + 12 = 0$.

c) State the discriminant of $2x^2 + 3x - 5 = 0$.